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വകുപ്പ്- കേരളം

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समग्र शिक्षा के रल
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ശ്രീ ബിനോയ് കുര്യൻ, വൈസ് പ്രസിഡണ്ട്, കണ്ണൂർ ജില്ലാ പഞ്ചായത്ത്
ശ്രീജിനി.എൻ വി, ചെയർപേഴ്സൺ ആരോഗ്യ - വിദ്യാഭ്യാസം, കണ്ണൂർ ജില്ലാ പഞ്ചായത്ത്

വിദഗ്ദ്ധസമിതി

ശ്രീ. വിനോദ് കുമാർ.കെ.എ., ആർ.ഡി.ഡി, കണ്ണൂർ
ശ്രീമതി.ഷൈനി.ഡി , ഡി.ഡി.ഇ. കണ്ണൂർ
ഡോ. വിനോദ് കുമാർ. കെ, പ്രിൻസിപ്പാൾ, ഡയറ്റ് കണ്ണൂർ
ശ്രീ. വിനോദ് ഇ.സി., ഡി.പി.സി, എസ്.എസ്.കെ., കണ്ണൂർ
ശ്രീമതി. ഉദയകുമാരി ഇ ആർ, അസിസ്റ്റന്റ് ഡയരക്ടർ, വി.എച്ച്.എസ്.ഇ.
ശ്രീ. കെ സുരേന്ദ്രൻ അടുത്തില, കോ - ഓർഡിനേറ്റർ, KITE കണ്ണൂർ

കോർഡിനേറ്റർമാർ

ഡോ. രാജേഷ് കെ പി, സീനിയർ ലക്ചറർ, ഡയറ്റ് കണ്ണൂർ
ഡോ: ദീപേഷ് കെ.വി., ജില്ലാ പ്രോഗ്രാം ഓഫീസർ, എസ്.എസ്.കെ. കണ്ണൂർ
ശ്രീമതി. സ്വാതി. വി, അസി.കോ - ഓർഡിനേറ്റർ (ഹയർ സെക്കണ്ടറി), കണ്ണൂർ
ശ്രീമതി. ബീന.കെ, ലക്ചറർ, ഡയറ്റ് കണ്ണൂർ

കവർ ഡിസൈൻ

ഷിബിൻ കെ.കെ.
GVHSS അമ്പലത്തറ, കാസറഗോഡ്

ജില്ലാവിദ്യാഭ്യാസപരിശീലനകേന്ദ്രം (ഡയറ്റ്), കണ്ണൂർ

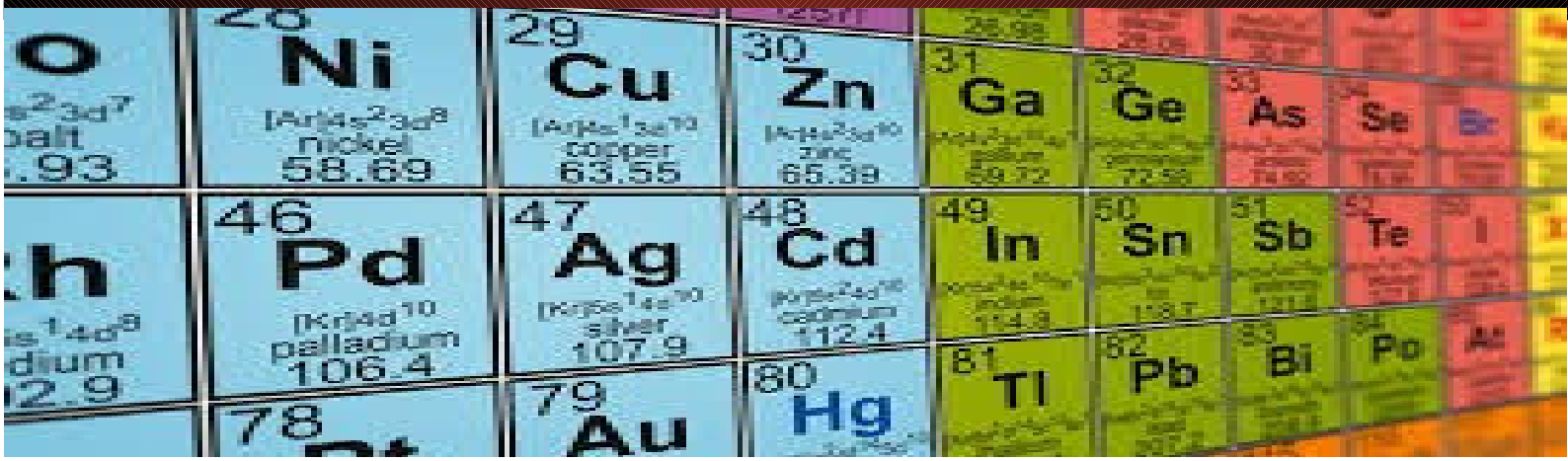
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CHEMISTRY



ജില്ലാ പഞ്ചായത്ത് - കണ്ണൂർ

ഡയറ്റ് കണ്ണൂർ

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കേവലം ചോദ്യങ്ങൾ എന്നതിലുപരി പഠന പ്രവർത്തനങ്ങൾക്ക് സാധ്യത നൽകാൻ രസതന്ത്രത്തിൽ ശ്രമിച്ചിട്ടുണ്ട്. ഭിന്ന നിലവാരത്തിലുള്ള കുട്ടികളെ പരിഗണിച്ചാണ് രസതന്ത്രം " SMILE " തയ്യാറാക്കിയിട്ടുള്ളത്. പുതിയ പാഠപുസ്തകത്തിന്റേയും ചോദ്യപേപ്പർ ഘടന മാറിയതിന്റേയും പ്രയാസങ്ങൾ ലഘൂകരിക്കാനുതകുന്ന പ്രവർത്തനങ്ങൾ നൽകിയിട്ടുണ്ട്. കണ്ണൂർ ഡയറ്റിന്റെ നിർദ്ദേശങ്ങളും അക്കാദമിക ഇടപെടലും ഏറെ ഉപയോഗപ്പെടുത്തിയിട്ടുണ്ട്.

രസതന്ത്രം ടെക്സ്റ്റ് ബുക്കിനൊപ്പം അദ്ധ്യാപകർക്കും കുട്ടികൾക്കും ഉപയോഗിക്കാൻ കഴിയുന്നതാണ് രസതന്ത്രം " SMILE" . കൂട്ടായ്മയിലൂടെ മുഴുവൻ കുട്ടികളെയും മികച്ച പഠന നിലവാരത്തിലേക്ക് എത്തിക്കേണ്ട ഉത്തരവാദിത്തം ഏറ്റെടുത്തുകൊണ്ട് ജില്ലാ പഞ്ചായത്തിന്റെ ഈ സദുദ്യമത്തെ പൂർണ്ണതയിലെത്തിക്കാൻ നമ്മുക്ക് സാധിക്കട്ടെ.

രസതന്ത്രം " SMILE" ശില്പശാലയിൽ പങ്കെടുത്തവർ

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എച്ച് എസ് ടി
മമ്പറം എച്ച് എസ് എസ്

കെ ഉണ്ണികൃഷ്ണൻ
എച്ച് എസ് ടി
കടമ്പൂർ എച്ച് എസ് എസ്

ഡോ: സൂരജ് എം പി
എച്ച് എസ് ടി
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CHEMISTRY

PART - I

Standard
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CHAPTER 1

NOMENCLATURE OF ORGANIC COMPOUNDS AND ISOMERISM

Nomenclature of alkanes with one branch

$$CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2 - CH_3$$

1) How many carbon atoms are present in the longest hydrocarbon chain ?
2) Number the longest carbon chain.
3) Find the position number of the branch?
4) Name the branching alkyl group
5) Write the IUPAC name

* Name of alkyl group = Word root corresponding to the number of carbon atom / atoms in the branch + 'yl'

* While writing the IUPAC name, a hyphen (–) is used to separate numerals and alphabets

Position number of branch+ hyphen + name of alkyl group + word root + suffix (ane)

$$CH_3 - CH_2 - CH_2 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$$

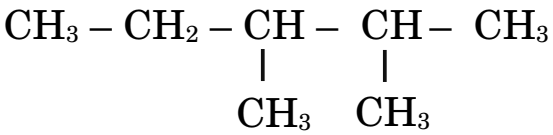
1) How many carbon atoms are present in the longest hydrocarbon chain
2) Number the longest carbon chain
3) Find the position number of the branch
4) Name the branching alkyl group
5) Write the IUPAC name

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Nomenclature of alkanes with more than one branch



1) How many carbon atoms are present in the longest hydrocarbon chain
2) Number the longest carbon chain
3) Find the position number of the branches
4) Name the branching alkyl groups
5) Write the IUPAC name

* If more than one branch is present, select the longest carbon chain. Number the carbon atoms from left to right or right to left in such a way that the carbon atoms with branches get the lowest position numbers.
* If the same branch appears more than once in a carbon chain, the number of branches is to be indicated using prefixes like di (two), tri (three), tetra (four) etc. Position numbers should be separated by commas.
* If the carbon atom containing the first branch gets the same position number when numbered from either side, then the numbering should be done in such a way that the carbon atom containing the second branch gets a lower position number.
* If a carbon atom has two identical branches, their position numbers should be repeated.

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Nomenclature of unsaturated hydrocarbons

In the nomenclature of hydrocarbons with double bond / triple bond , the numbering should be done in such a way that the carbon atoms linked by the double bond / triple bond gets the lowest position number.

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For alkene

*Word root + hyphen + position of double bond + hyphen + suffix (ene)

$CH_3 – CH_2 – CH_2 – CH = CH_2$ (Pent- 1- ene)

$CH_3 – CH = CH – CH_2 – CH_3$ (.....)

For alkyne

*Word root + hyphen + position of triple bond + hyphen + suffix (yne)

$CH_3 – CH_2 – CH_2 – C \equiv CH$ (Pent- 1- yne)

$CH_3 – CH_2 – C \equiv C – CH_2 – CH_3$ (.....)

Functional groups

An atom or a group of atoms, bonded to carbon in an organic compound, determines the distinctive chemical and physical properties of that compound. This atom or group of atoms is called a functional group.

1. Hydroxyl group (–OH)

According to IUPAC method, the alcohols are named by replacing 'e' in the name of the corresponding alkane with 'ol'.

Alkane - e + ol →Alkanol

Eg. $CH_3 – OH$ Methane - e + ol → Methanol

$CH_3 – CH – OH$ IUPAC name .....

When there are more than two carbon atoms, the position of the –OH group needs to be specified. For naming such compounds according to IUPAC rules, the following points need to be considered.

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*The carbon chain containing the –OH group should be considered as the main chain.

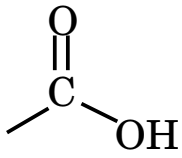
*The carbon atoms should be numbered in such a way that the carbon to which the functional group is attached gets the lowest position number.

*Replace 'e' of the corresponding alkane with 'ol' and indicate the position number of the –OH group before 'ol'.

Alkane – e + hyphen + position number of –OH group +hyphen + ol

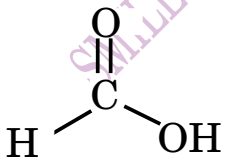
$CH_3 – CH_2 – CH_2 – CH_2 – OH$ (butan – 1- ol)

$CH_3 – \underset{\displaystyle OH}{\underset{|}{CH}} – CH_2 – CH_3$ (.....)

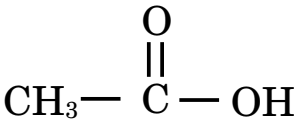
2. <u>Carboxyl group</u> – COOH or 

The last letter 'e' of the corresponding alkane is replaced with 'oic acid' to get the name of the acid.

Alkane – e + oic acid → Alkanoic acid

Eg.  or H – COOH

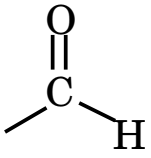
Methane – e + oic acid → Methanoic acid

 or $CH_3 – COOH$ IUPAC name .....

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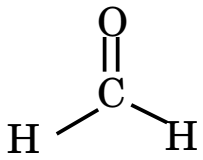
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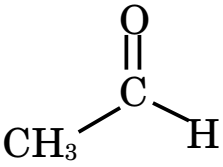
3. <u>Aldehyde group</u> – CHO or 

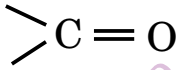
The last letter 'e' in the name of the corresponding alkane is replaced with '–al' to get the name of the aldehyde.

Alkane – e + al → Alkanal

Eg.  or H - CHO

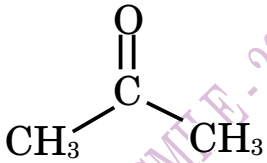
Methane – e + al → Methanal

 or CH_3 - CHO IUPAC name .....

4. <u>Keto group</u> 

Here the letter 'e' of the corresponding alkane is replaced with 'one'.

Alkane – e + one → Alkanone

 or CH_3 - CO - CH_3 propanone (Acetone)

The position of the functional group must be considered while naming ketones with more than 3 carbon atoms.

CH_3 - CH_2 - CH_2 - CO - CH_3 Pentan- 2- one

CH_3 - CH_2 - CO- CH_2 - CH_3 IUPAC name .....

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5. Halo group (–F, –Cl, –Br, –I)

The compounds formed when one or more hydrogen atoms in a hydrocarbon is replaced with an equal number of halogen atoms are called halo compounds. There are organic compounds with functional groups such as fluoro (–F), chloro (–Cl), bromo (–Br), and iodo (–I).

The IUPAC nomenclature of halo compounds with more than two carbon atoms in the main chain is given below.

Position of the halo group +hyphen + name of the halo group + name of the alkane

$CH_3 - CH_2 - CH_2 - Cl$ 1- chloro propane

$CH_3 - \underset{\displaystyle Cl}{\underset{|}{CH}} - CH_2 - CH_3$ IUPAC name .....

$CH_3 - \overset{\displaystyle Br}{\overset{|}{\underset{\displaystyle Br}{\underset{|}{C}}}} - CH_2 - CH_3$ IUPAC name .....

6. Alkoxy group (– O – R)

*Ethers are compounds containing alkoxy group.
‘ R ’ denotes an alkyl group
*Ethers are named as alkoxyalkane.
* – O – group is called ether linkage. Of the alkyl groups on either side of ether linkage (–O–), the longer alkyl group is considered as alkane and the shorter as alkoxy group.

Eg. $CH_3 - CH_2 - O - CH_3$ methoxyethane

$CH_3 - CH_2 - O - CH_2 - CH_2 - CH_3$ IUPAC name .....

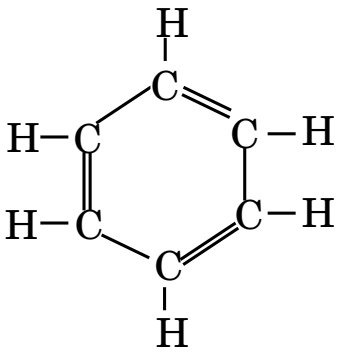
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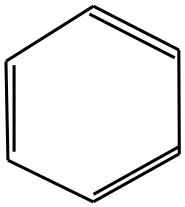
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Aromatic compounds

Eg. benzene (C_6H_6)

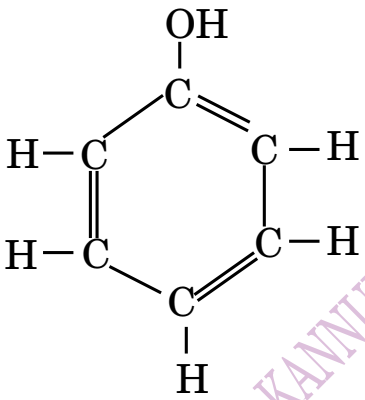


OR

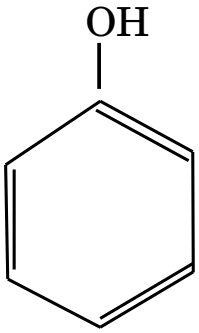


Phenol ($C_6H_5 - OH$)

Phenol is the compound obtained when a hydrogen atom in benzene is replaced with an –OH group.

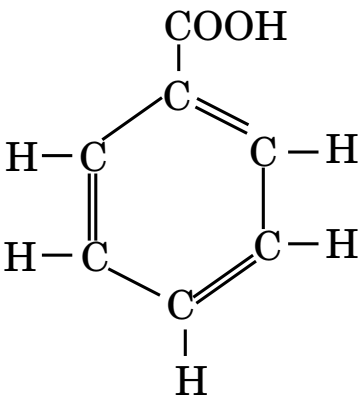


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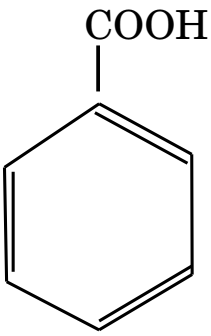


Benzoic acid ($C_6H_5 - COOH$)

Benzoic acid is the compound obtained when a hydrogen atom in benzene is replaced with an –COOH group.



OR



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Isomerism

Compounds having same molecular formula and different chemical and physical properties are called isomers. This phenomenon is called isomerism. The structural formulae of these compounds are different.

Chain isomerism

The structural formulae of two compounds are given below.

i) $CH_3 - CH_2 - CH_2 - CH_3$

ii) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$

Write the molecular formulae of these two compounds .....

What is the peculiarity in the molecular formulae? .....

Write their IUPAC names.

i) .....

ii) .....

Compounds that have the same molecular formula but differ in the structures of carbon chain are called chain isomers. This phenomenon is known as chain isomerism.

Position isomerism

The structural formulae of two familiar compounds are provided below.

i) $CH_3 - CH_2 - CH_2 - OH$

ii) $CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_3$

What is the functional group present in them? .....

What is their molecular formula? .....

Write their IUPAC name.

Compound (i) .....

Compound (ii) .....

When two compounds have same molecular formula and same functional group, but differ in the position of the functional group, they are called position isomers, and this phenomenon is known as position isomerism.

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More examples for Position Isomerism:-

i) $CH_3 - CH_2 - CH_2 - OH$ & $CH_3 - \underset{\underset{OH}{|}}{CH} - CH_3$

ii) $CH_3 - CH_2 - CH_2 - Cl$ & $CH_3 - \underset{\underset{Cl}{|}}{CH} - CH_3$

iii) $CH_3 - CH_2 - CH_2 - CH = CH_2$

& $CH_3 - CH = CH - CH_2 - CH_3$

iv) $CH_3 - CH_2 - CH_2 - C \equiv CH$

& $CH_3 - C \equiv C - CH_2 - CH_3$

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<u>Functional isomerism</u>

i) $CH_3 - CH_2 - OH$

ii) $CH_3 - O - CH_3$

What is the functional group present in each compound?
Compound (i) ....., Compound (ii) .....
Write their molecular formula. .....

When compounds have the same molecular formula but different functional groups, they are known as functional isomers, and this phenomenon is called functional isomerism.

Eg. for functional isomerism

$CH_3 - CH_2 - CHO$ & $CH_3 - CO - CH_3$

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Metamerism

Analyse the structural formulae of two ethers given below.

i) $CH_3 – CH_2 – O – CH_2 – CH_3$

ii) $CH_3 – O – CH_2 – CH_2 – CH_3$

What is their molecular formula? .....

Write their IUPAC name.

Compound (i) .....

Compound (ii) .....

Compounds having same molecular formula but different alkyl groups attached to either side of the functional group are called metamers. The isomerism exhibited by compounds with the same molecular formula but different alkyl groups on either side of the bivalent functional group (group having valency 2, eg.(- O - , $>C = O$) is known as metamerism

Eg:- $CH_3 - CH_2 - CH_2 - CO - CH_3$

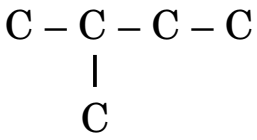
$CH_3 - CH_2 - CO - CH_2 - CH_3$

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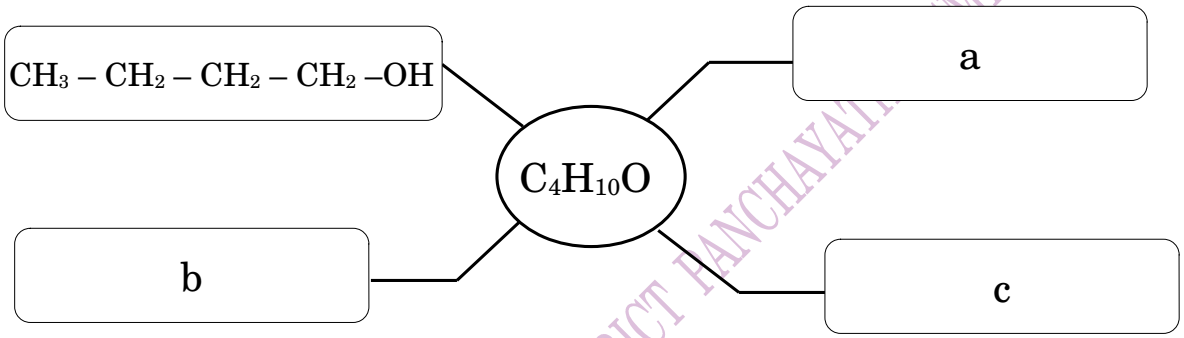
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1. Incomplete structure of a aliphatic hydrocarbon is given below.



a) Complete the structure by adding hydrogen atoms.
b) Write the molecular formula of this compound.
c) Write the structure of all possible isomers of this compound.

2. a , b , c are different isomers of $C_4H_{10}O$.



i) Write the structure of a , b, and c.

ii) Write a pair of functional isomers among them.

3. What is the IUPAC name of an alkane with total of 4 carbon atoms and has a methyl group in the second carbon atom of main chain.?

a) 2- methyl butane
b) 4- methyl butane
c) 2- methyl propane
d) 4- methyl propane

4. What is the total number of carbon atoms in 2,3- dimethyl pentane?

(5 , 6 , 7, 10)

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5. $CH_3 - CH_2 - \underset{\underset{O}{\|}}{C} - OH$

a) Name the functional group in the above compound?

b) Write the IUPAC name of this compound ?

6. A $CH_3 - CH_2 - \underset{\underset{O}{\|}}{C} - OH$

B $CH_3 - \underset{\underset{O}{\|}}{C} - CH_3$

C $CH_3 - CH_2 - \underset{\underset{O}{\|}}{C} - H$

C $CH_3 - CH_2 - CH_2 - OH$

a) Select the isomer pair from the above list.

b) Among these , which is carboxylic acid?

c) Which compound contain keto group?

d) Write the IUPAC name of aldehyde from the above list?

7. Find isomer pairs from the compounds given below and write the type of isomerism.

A $CH_3 - \underset{\underset{OH}{|}}{CH} - CH_3$

E $CH_3 - \underset{\underset{CH_3}{|}}{CH} - CH_3$

B $CH_3 - CH_2 - CH_2 - O - CH_3$

F $CH_3 - CO - CH_2 - CH_3$

C $CH_3 - CH_2 - CH_2 - CH_3$

G $CH_3 - CH_2 - CH_2 - CHO$

D $CH_3 - CH_2 - CH_2 - OH$

H $CH_3 - CH_2 - O - CH_2 - CH_3$

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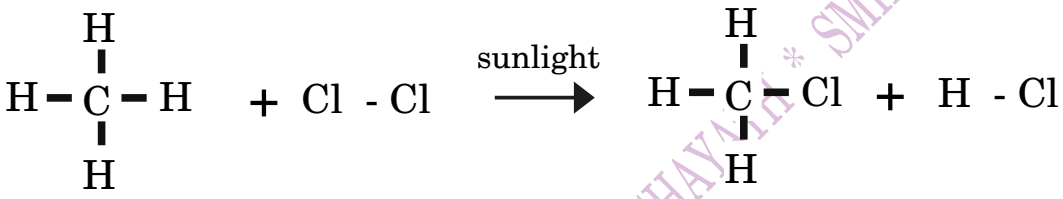
CHAPTER 2

CHEMICAL REACTIONS OF ORGANIC COMPOUNDS

1. Substitution reactions

Substitution reactions are chemical reactions in which an atom or group of atoms is replaced with another atom or group of atoms in a compound.

Eg: Reaction of methane with chlorine in the presence of sunlight.



CH_4 + Cl_2 $\xrightarrow{\text{sunlight}}$ CH_3Cl + HCl

methane chlorine chloro methane

2. Addition reactions

Addition reactions are reactions in which unsaturated organic compounds with double or triple bonds combine with certain molecules to form saturated compounds. The reactions in which triple bonded organic compounds partially combine with small molecules to form double bonded compounds are also addition reactions.

Reactants	Product
$CH_2 = CH_2 + H_2$	
$CH_2 = CH_2 +$	$CH_3 — CH_2Cl$
$CH \equiv CH + Cl_2$	
$CH \equiv CH + 2H_2$	

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3. Polymerisation

Polymerisation is the process by which simple molecules join together to form large complex molecules.

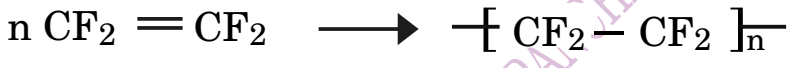
The simple molecules are called monomers and the large complex molecules formed as a result of polymerisation are called polymers.

a) Addition polymers

Addition polymers are obtained by the repeated addition reaction of monomers.



Ethene polyethene (polythene)



Tetrafluoroethene Polytetrafluoroethene (Teflon)

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Monomer	Polymer	Uses
Vinyl chloride	PVC	Manufacture of pipes, plastic furniture, coating of electric conductors etc.
Ethene	Polythene	Manufacture of tarpaulin sheets, carry bags etc.
Isoprene	Natural rubber	Manufacture of tyres.
Tetrafluoroethene	Teflon	Coating of the inner surface of non stick cookware.
Acrylonitrile (Vinyl cyanide)	Orlon	Manufacture of synthetic fibres.

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b) Condensation polymers

Condensation polymerisation is the process in which different monomers combine together to form larger compounds accompanied by the removal of simple molecules.

$$n\text{HOOC-}(CH_2)_4\text{-COOH} + n\text{NH}_2\text{-}(CH_2)_6\text{-NH}_2 \longrightarrow$$

Adipic acid Hexamethylenediamine

$$\left[\text{OC-}(CH_2)_4\text{-CO-NH-}(CH_2)_6\text{-NH}\right]_n + n\text{H}_2\text{O}$$

Nylon 66

Monomer	Polymer	Uses
Adipic acid & Hexamethylenediamine	Nylon 66	Manufacture of fabrics, combs, bristles of brushes etc.
Phenol & Formaldehyde	Phenol formaldehyde resin (bakelite)	Manufacture of switches, plugs, handles of pressure cookers etc.
Ethylene glycol & Terephthalic acid	Polyethylene terephthalate (polyester)	Manufacture of tarpaulin, bottles, fabrics etc.

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4. Thermal cracking

When heated in the absence of air, some hydrocarbons with high molecular weight decompose into hydrocarbons with lower molecular weight. This process is called thermal cracking.

$$CH_3 - CH_2 - CH_3 \xrightarrow{\text{heat}} CH_4 + CH_2 = CH_2$$

$$CH_3 - CH_2 - CH_2 - CH_3 \xrightarrow{\text{heat}} CH = CH_2 + \text{.....}$$

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5. Combustion of hydrocarbon

When hydrocarbons burn, they combine with oxygen in air to form CO_2 and H_2O along with heat and light. This process is called combustion.

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + \text{Heat}$$

$$2C_4H_{10} + 13O_2 \longrightarrow 8CO_2 + 10\,H_2O + \text{Heat}$$

Methanol ($CH_3 - OH$) – wood spirit

Methanol is industrially produced by treating carbon monoxide with hydrogen in the presence of catalysts.

$$CO + 2H_2 \xrightarrow[573\ K]{\text{catalyst}} CH_3 - OH$$

Uses of methanol

* For manufacturing varnish, paint etc.
* For the manufacture of formic acid, formaldehyde etc.

40% solution of formaldehyde is formalin.

Ethanol ($CH_3 - CH_2 - OH$)

Ethanol is manufactured by the fermentation of molasses. Yeast is added to the dilute molasses. The enzyme invertase present in yeast converts sugar solution to glucose and fructose. Then another enzyme zymase, converts glucose and fructose into ethanol.

$$\underset{\text{Sugar solution}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{\text{Invertase}} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

$$\underset{\text{Glucose/ Fructose}}{C_6H_{12}O_6} \xrightarrow{\text{Zymase}} \underset{\text{ethanol}}{2C_2H_5\text{- }OH} + 2CO_2$$

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wash :- 8-10% ethanol is known as wash. It is obtained by the fermentation of molasses.

Rectified spirit:- 95.6% ethanol is known as rectified spirit. It is obtanied by the fractional distillation of wash.

Absolute alcohol:- 100% of ethanol is known as absolute alcohol.

Power alcohol:-Power alcohol is a mixture of 20% absolute alcohol and 80% of petrol. It is used as fuel in vehicles.

Denatured spirit:- Toxic substances like methanol/pyridine/rubber distillate etc. are added to ethanol in order to prevent its misuse as a beverage. Ethanol thus obtained is called denatured spirit.

Methylated spirit:- Ethanol with toxic substance methanol is called methylated spirit.

Uses of ethanol:-

* Production of power alcohol.
* As a solvent for medicines.
* Manufacture of paints.
* As preservatives.
* Production of organic compounds.

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Ethanoic acid ($CH_3 - COOH$) - acetic acid

Ethanoic acid can be prepared industrially by treating methanol with carbon monoxide in the presence of a catalyst.

$$CH_3 - OH + CO \xrightarrow{\text{catalyst}} CH_3 - COOH$$

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Fermentation of ethanol with acetobacter bacteria in the presence of air yields less concentrated (5-8%) ethanoic acid. This is called vinegar.

Uses of ethanoic acid

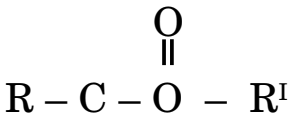
* Manufacture of vinegar.
* Production of acetic anhydride, acetate ester,synthetic fibres etc.
*Solvent of polymers and resins.
* Manufacture of disinfectants.
* Manufacture of medicines.

Esters

Esters are formed when alcohols react with carboxylic acids. This reaction is called esterification. As esters have the fragrance of flowers and fruits, they are used to make artificial perfumes and juices.

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The general formula of esters is



$$CH_3 - CH_2 - COOH + HO - CH_3 \xrightarrow{H_2SO_4} CH_3 - CH_2 - COO - CH_3 + H_2O$$

Propanoic acid — methanol — Methyl propanoate

$$CH_3 - COOH + HO - CH_2 - CH_2 - CH_3 \xrightarrow{H_2SO_4} \text{.....} + H_2O$$

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Name of carboxylic acid	Name of alcohol	Name of ester	Fragrance
Ethanoic acid	Isoamyl alcohol	Isoamyl acetate	Banana
Ethanoic acid	Benzyl alcohol	Benzyl ethanoate	Jasmine
Ethanoic acid	Octyl alcohol	Octyl ethanoate	Orange
Butanoic acid	Ethyl alcohol	Ethyl butanoate	Pineapple

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Medicines

The category and functions of certain medicines used in modern methods of medical therapy are given below.

Category	Function	Examples
Analgesics	Relieve pain	Aspirin
Antipyretics	Reduce body temperature	Paracetamol
Antiseptics	Control microorganism	Dettol
Antibiotics	Destroy infectious microorganisms and prevent their growth.	Penicillin

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1. a) Complete the chemical equations given below.

i) $C_2H_4 + Cl_2 \longrightarrow$ .....

ii) ..... + $Cl_2 \longrightarrow CH_2Cl_2 + HCl$

b) Write the type of chemical reactions

Reaction i) .....

Reaction ii) .....

2. Features of an organic compound are given below.

i) It has two carbon atoms.

ii) Double bond is present between carbon atoms.

iii) Functional group in this compound is 'chloro'

a) Write the structure of this compound.

b) Write the IUPAC name of this compound.

c) Name the polymer formed when this compound undergone addition polymerisation.

3. Which of the following is condensation polymer?

a. Polyvinyl chloride b. Nylon66 c. Polythene d. Teflon

4. $CO + 2H_2 \xrightarrow[573\ K]{\text{catalyst}} A$

$A + CO \xrightarrow{\text{catalyst}} B$

$A + B \xrightarrow{\text{Conc. } H_2SO_4} C + H2O$

a) Write the condensed formula of a,b and c

b) What is the general name to which 'C' belongs?

c) Write the IUPAC name of 'C" .

5. Ethanol is an industrially important compound.

a) What is the name of 8 – 10 % of ethanol ?

b) How it is converted in to rectified spirit ?

c) What is denatured spirit ?

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6. Complete the table

Reactants	Products	Name of reaction
CH_4 + Cl_2	..... + HCl	.....
$CH_3 - CH_2 - CH_2 - CH_3$	CH_4 + ...	.....
C_2H_4 + ...	C_3H_6	Addition reaction
CH_4 + O_2	..... + ...	combustion
n $CH_2 = CH_2$	.....	.....
$CH_3 - COOH$ + CH_3 - OH	..... + H_2O	.....

7. a) How many hydrogen molecules are required to convert propyne into propane ?
   b) Write the chemical equation of the reaction.
   c) To which category does this chemical reaction belong?

8. Three chemical equations are given below.

i) $CH_3 - CH_2 - CH_2 - CH_3 \xrightarrow{Heat} CH_3 - CH_3 + P$

ii) $P + H_2 \xrightarrow{Ni} Q$

iii) $Q + Cl_2 \xrightarrow{Sunlight} R + HCl$

a) Write the structure of 'P' , 'Q' and 'R'

b) Write the name of chemical reaction (ii) and (iii)

c) What is the IUPAC name of 'R' ?

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CHAPTER 3

PERIODIC TABLE AND ELECTRON CONFIGURATION

According to Bohr model of atom, electrons revolve around the nucleus of an atom in fixed orbits. Orbits have definite energy and hence they are known as stationary energy levels. But, later studies showed that Bohr model of atom has certain limitations.

Louis de Broglie discovered the wave nature of matter. The scientific world has accepted that electrons exhibit both particle and wave properties.

According to Heisenberg's Uncertainty Principle, it is impossible to determine simultaneously the exact position and the exact velocity of a fast moving subatomic particle like electron.

An atom model that is in line with the dual nature of matter and the uncertainty principle was developed on the basis of quantum mechanics..

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Orbitals

The three dimensional regions around the nucleus where there is maximum probability of finding the electrons are called orbitals.

The numbers which are used to describe the characteristics of orbitals and electrons based on quantum mechanical model of atom are called quantum numbers.

1. Principal Quantum Numbers (n)

Principal quantum number (n) is used to represent the shells or principal energy levels.

Principal quantum number (n)	1	2	3	4
Shell	K	L	M	N

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2. Azimuthal Quantum Number (l)

Azimuthal quantum number (l) defines the three dimensional shape of the orbital. Azimuthal quantum number is also used to represent the subshells of each shell. The subshells are denoted using the symbols s, p, d and f.

The value of l = zero to (n – 1)

Eg: Value of l for N Shell = zero to (n – 1)

l = zero to (4 – 1) = 0,1,2,3

N shell have 4 sub shells. Their values are 0,1,2,3.

Complete the table

Principal quantum number (n)	1	2		3			4			
Shell	K	L		M			N			
l							0	1	2	3
Sub Shell							s	p	d	f

3. Magnetic Quantum Number (m)

The quantum number that represents the difference in the orientation of orbitals is called magnetic quantum number.

Value of m = $(2l + 1)$

Eg:

s sub shell

$l = 0$

$m = 2l + 1$

$= 2 \times 0 + 1$

$= 1$

1 orientation.

d sub shell

$l = 2$

$m = 2l + 1$

$= 2 \times 2 + 1$

$= 5$

5 orientation

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Complete the table

Principal quantum number (n)	1	2		3			4			
shell	K	L		M			N			
sub shell	s	s	p	s	p	d	s	p	d	f
Number of orbitals in sub shell	1					5			5	
Total number of orbitals in shell	1									

Number of orbitals that can be accommodated in each shell = n^2
Number of electrons that can be accommodated in each shell = $2n^2$
Number of electrons that can be accommodated in each orbital = 2
Number of electrons that can be accommodated in each subshell = 2 (2 l + 1)
s - 2 , p - 6 , d - 10 , f - 14

Filling of electrons in subshells

When the electrons in an atom are distributed in subshells, they are being filled gradually in the increasing order of energies of the subshells. This is called subshell electron configuration.

The order of energy can be found from the principal quantum number (n) and azimuthal quantum number (l) values representing each subshell.

The energy of subshells increases in the ascending order of (n + l) values. If (n + l) values of subshells are the same , it is considered as the subshell with higher " n " value has more energy. Subshells in the ascending order of energy are given below.

1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s ....

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Complete the table

Element	subshell electron configuration
$_{4}$Be	
$_{8}$O	
$_{11}$Na	
$_{18}$Ar	
$_{21}$Sc	
$_{30}$Zn	

Block from subshell electron configuration

The subshell to which the last electron is added will be the same as the block to which the element belongs.

Complete the table.

Element	subshell electron configuration	Block
$_{3}$Li		
$_{9}$F		
$_{26}$Fe		

Period number from subshell electron configuration

The period number of an element is the highest shell number in its subshell electron configuration.

period number = highest shell number

Complete the table.

Element	subshell electron configuration	highest shell number	period number
$_{6}$C			
$_{19}$K			
$_{22}$Ti			

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Group number from subshell electron configuration

group number of s block elements = the number of electrons in the outermost s subshell

Complete the table.

Element	subshell electron configuration	Block	Group number
$_{20}Ca$			
$_{12}Mg$			
$_{4}Be$			

group number of p block elements = total number of electrons in the outermost s and p subshells + 10

Complete the table.

Element	subshell electron configuration	Block	Group number
$_{6}C$			
$_{16}S$			
$_{31}Ga$			

group number of d block elements = number of electrons in the outermost s subshell + number of electrons in the penultimate d subshell

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Complete the table.

Element	subshell electron configuration	Block	Group number
$_{23}V$			
$_{28}Ni$			
$_{25}Mn$			

Periodic trend in periodic table

Ionisation enthalpy

Ionisation enthalpy of an element is the minimum amount of energy required to remove the most loosely bound electron from the outermost shell of an isolated gaseous atom of the element.

* Ionisation enthalpy decreases on moving down a group.

* On moving from left to right in a period ionisation enthalpy increases.

Characteristics of S block elements

* When these elements undergo chemical reactions, they donate their outermost s electrons.
* The elements of group 1 and group 2 exhibit +1 and +2 oxidation states respectively.
* s block elements generally exist in solid state.
* Caesium ($_{55}Cs$) exists in liquid state on warm days. (melting point of $_{55}Cs$ is 28.40^0C)
* The elements Francium ($_{87}Fr$) and Radium ($_{88}Ra$) are radioactive in nature.

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Characteristics of p block elements

* p block elements include metals, non metals and metalloids. These elements exist in solid, liquid and gaseous states.
* p block elements exhibit both positive (+) and negative (–) oxidation states.
* Gallium exists in liquid state on warm days. (melting point of Gallium is 29.77^0C)

Characteristics of d block elements

* The electrons are being gradually filled up in the penultimate shell.
* All the d block elements are metals.
* They show similarities in properties not only within the groups but also along the periods.
* They show variable oxidation states.
* The compounds of transition elements are generally coloured.

Characteristics of f block elements

* filling up of electrons takes place in the anti penultimate shell.
* They show variable oxidation states.
* Actinoids are radioactive elements. These include man-made elements as well.
* The elements after uranium are man-made.

Uses of f block elements

* Certain isotopes of elements like Uranium (U), Thorium (Th) and Plutonium (Pu) are used as fuel in nuclear reactors.
* Neodymium (Nd) is used for making strong magnets.
* Some elements are used as catalyst in the petroleum industry. Eg:- Cerium (Ce), Lanthanum (La).

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1. Which subshell is represented by the quantum number values given below?

   n = 3 , l = 2

   (a) 3s (b) 2p (c) 3d (d) 3p

2. Pairs of quantum number values of some subshells are given below.

   a) n = 2 , l = 1

   b) n = 2 , l = 0

   c) n = 2 , l = 2

   Choose the correct option from the following.

   i) a , b ,c are correct

   ii) a is wrong , b & c are correct

   iii) a & b are wrong , c is correct

   iv) a & b are correct , c is wrong

3. Select the correct order of increasing energy of the subshells from the following.

   (i) 6s < 5p < 5d < 4f

   (ii) 4f < 5p < 5d < 6s

   (iii) 5p < 6s < 5d < 4f

   (iv) 5p < 6s < 4f < 5d

4. * Electron configuration of an element ends in a subshell with the quantum numbers n = 3 , l = 2

   * 5 electrons are present in this subshell.

   a) Write subshell electron configuration of this element.

   b) To which block does this element belongs?

5. A portion of periodic table is given below. Symbols are not real

A	B	C
D	E	F

Electron configuration of " D " is [Ar] $3d^{10}$ $4s^2$ $4p^3$.

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a) Write block, period and group number of element " D " .
b) What is the atomic number of element " C " ?

6. Select the correct electron configuration of $_{24}Cr$-from the options given below.
   (i) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$
   (ii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^4\ 4s^2$
   (iii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$

7. The element "A" belong to second period and 17th group and the element "B" belong third period and second group of the periodic table.(Symbols shown are not real)
   a .Write the subshell electron configuration of "A"
   b . To which block does "B" belong? What is its valency?
   c .Write the chemical formula of the compound formed by combining "A" and "B"

8. Atomic number of the element of " X " is 25. Symbol is not real. The oxides of " X " are X_2O_3 and X_2O_5
   a . Write the subshell electron configuration of X?
   b. What is the oxidation state of "X" in X_2O_5?
   c . Write the subshell electron configuration of X^{+3} ion?

9. Complete the table given below.

Element	[Xe] $4f^{14}\ 6s^2$	[Ne] $3s^2\ 3p^6$	[Ar] $3d^{10}\ 4s^2$
Block			
Period Number			

10. Total number of "s" electrons of an element is 7.
    a) Write subshell electron configuration of this element.
    b) Find out block , period and group of this element ?

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11. Electron configuration of some elements are given.

(Symbols given are not real)

A -$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^5$

B -$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^1$

C -$1s^2$ $2s^2$ $2p^6$ $3s^1$

D -$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$

a) Element "C" has biggest atom.

b) Element "D" does not take part in chemical reaction.

c) Element "A" shows positive oxidation state in their compounds.

Examine the statements given above and select the correct option.

i) a , b ,c are correct

ii) a is wrong , b and c are correct

iii) a & b are wrong , c is correct

iv) a & b are correct , c is wrong

12. Sub shell electronic configuration of X^{2-} ion is $1s^2$ $2s^2$ $2p^6$.

X is not a real symbol.

a) Write subshell electronic configuration of " X "

b) Write chemical formula of compound formed when element " X " combine with element " Y " belong to second group of the periodic table.

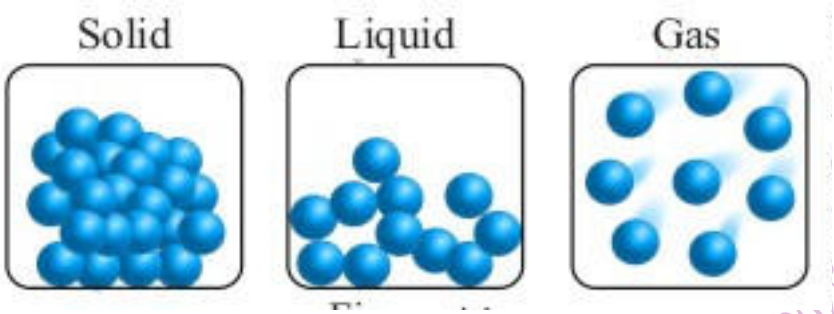
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CHAPTER 4

GAS LAWS AND MOLE CONCEPT



Compare the distance between the molecules, force of attraction between them, freedom of movement of the molecules and their energy in gaseous state, with those in the other states of matter and write down.

Distance between the molecules	Very high
Force of attraction between the molecules	
Freedom of movement of the molecules	
Energy of the molecules	

The postulates of kinetic molecular theory proposed by James Clerk Maxwell and Ludwig Boltzmann to explain these properties are given below.

a) Gases are made up of minute particles (atoms/molecules).

b) The attractive force between gaseous molecules is very low.

c) As the molecules are so far apart, the volume of gaseous molecules is negligible in comparison with the total volume of the gas.

d) The volume can be reduced by reducing the distance between the gaseous molecules.

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e) Gaseous molecules are in constant motion in all directions. As a result, the molecules collide with one another and with the walls of the container. The force produced due to the collision of molecules with the walls of the container results in gaseous pressure.

f) Collisions of gaseous molecules are elastic in nature. That means, the kinetic energy of the molecules before and after the collision will be same.

g) The average kinetic energy of the gaseous molecules is directly proportional to its temperature.

General properties of gases

Volume

The volume of a gas is the volume of the container in which it is occupied. The SI unit of volume is m^3

Pressure

Gaseous molecules are in constant random motion. As a result, they collide with one another and with the walls of the container. Due to these collisions, a force is experienced on the surface.

Gaseous pressure is the force experienced per unit surface area.

$$\text{Pressure} = \frac{\text{Force}}{\text{Surfce area}} \qquad P = \frac{F}{A}$$

The SI unit of pressure is Pascal (Pa)

Temperature

When gases are heated, the energy of molecules increases and hence temperature also increases.

The SI unit of temperature is Kelvin(K). To convert the common unit ^0C to Kelvin, add 273 to it.

$$t°C = (t + 273)\ K$$

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Gas Laws

Boyle's law

At constant temperature, the volume of a fixed mass of gas is inversely proportional to its pressure.

That is, $V \alpha \frac{1}{P}$ (Temperature, mass constant)

V = a constant $\times$ $1 / P$

$PV = k$ (k = a constant)

If the volume of a fixed mass of gas is $\mathbf{V_1}$ at pressure $\mathbf{P_1}$ and $\mathbf{V_2}$ at pressure $\mathbf{P_2}$, Then

$$\mathbf{P_1V_1 = P_2V_2}$$

The pressure and volume of a given mass of gas at constant temperature are given in the table. Complete the table.

P (atm)	V (L)
2	30
4	.....
.....	8
3	.....
5	

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Charles's Law

At constant pressure, the volume of a definite mass of a gas is directly proportional to its temperature in Kelvin Scale. This is Charles's law.

$V \alpha T$ (pressure, mass constant)

$V = k \times T$ (k - a constant)

$$\frac{V_1}{T_1} = k, \text{ a constant}$$

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If the volumes of a definite mass of gas are $\mathbf{V_1}$ and $\mathbf{V_2}$ at temperatures T_1 and $\mathbf{T_2}$ respectively, then

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

The volume of a given mass of gas at constant pressure and its temperature in Kelvin scale are given in the table. Complete the table.

T (K)	V (mL)	V / T
200 K	600	.....
300 K	.....	3
.....	300	.....
500 K	.....	.....

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Avogadro's Law

At constant temperature and pressure, the volume of a gas is directly proportional to the number of molecules. This is Avogadro's Law.

$V \propto N$ (Temperature, pressure constant)

Combined gas equation

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Mole concept

A mole is the quantity of a substance containing 6.022×10^{23} particles (atoms / molecules / ions).This number came to be known as Avogadro number (N_A) named in honour of the Italian scientist Amedeo Avogadro.

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Mole is the SI unit of quantity of matter. This represents the number of particles in that substance.

Relative atomic mass and mole

Relative atomic mass expresses the mass of one atom relative to that of the another .It indicates how many times one atom is heavier as compared to another.

Element	Average atomic mass	Relative atomic mass
Hydrogen	1.0079	1
Oxygen	15.9994	16
Sodium	22.989	23
Carbon	12.011	12
Nitrogen	14.0067	14

An element that weighs as much as its relative atomic mass in grams contains 6.022×10^{23} atoms. This mass is known as gram atomic mass. One gram atomic mass contains 1 mole atoms.

Element	Relative atomic mass	Gram atomic mass	Number of mole atoms	Number of atoms
Copper	63.5	63.5 g	1	6.022×10^{23}
Iron	55.8	55.8 g	1	6.022×10^{23}
Zinc	65.3	65.3 g	1	6.022×10^{23}
Aluminium	27	27 g	1	6.022×10^{23}
Nitrogen	14	14 g	1	6.022×10^{23}

One gram atomic mass ⟷ 1 mole atom ⟷ 6.022×10^{23} atoms

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Element	Gram atomic mas	Given mass	Number of Mole atom	Number of atoms
Aluminium	27 g	27 g	1	6.022×10^{23}
Aluminium	27 g	54 g	2	2 x 6.022×10^{23}
Aluminium	27 g	270 g	.....	.....
Aluminium	27 g	.....	5	.....
Aluminium	27 g			3 x 6.022×10^{23}

$$\text{Number of mole atom} = \frac{\text{Given mass}}{\text{Gram atomic mass}}$$

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Molar Mass

The total mass of atoms in a molecule is known as molecular mass.

(Atomic mass - H = 1, O = 16, N = 14, C = 12, S = 32)

Element / Compound	Chemical formula	Molecular mass
Oxygen	O_2	2 × 16= 32
Ammonia	NH_3	14 + 3 × 1 = 17
Water	H_2O	.....
Glucose	$C_6H_{12}O_6$	.....
Sulphuric acid	H_2SO_4	.....
Nitrogen	N_2	.....

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The molecular mass of an element / compound expressed in grams is known as gram molecular mass.

Mass of 6.022×10^{23} molecule is one gram molecular mass of that molecule. This mass is known as molar mass.

One molar mass of a compound contains one mole molecules.

Substance	Molar mass	Given mass	Number of Moles	Number of molecules
Water (H_2O)	18 g	18 g	1	6.022×10^{23}
Water (H_2O)	18 g	36 g	2	2 x 6.022×10^{23}
Water (H_2O)	18 g	360 g	.....	.....
Water (H_2O)	18 g	.....	5	.....
Water (H_2O)	18 g			3 x 6.022×10^{23}

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$$\text{Number of moles} = \frac{\text{Given mass}}{\text{Molar mas}}$$

(Atomic mass - H = 1, O = 16, N = 14, C = 12, S = 32, Ca = 40, Na = 23, Cl = 35.5)

Substance	Molar Mass	Given mass	Number of moles	Number of molecules
NH_3	.....	85 g	.....	.... $\times 6.022 \times 10^{23}$
CO_2	.....	220 g	.....	.....
H_2O	.....	.....	10	$10 \times 6.022 \times 10^{23}$
NO_2	.....	92 g	.....	.....
$C_6H_{12}O_6$	.....	360 g	.....	.....
NaCl	58.5 g	1170 g	.....	.....

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Volume of gases and mole

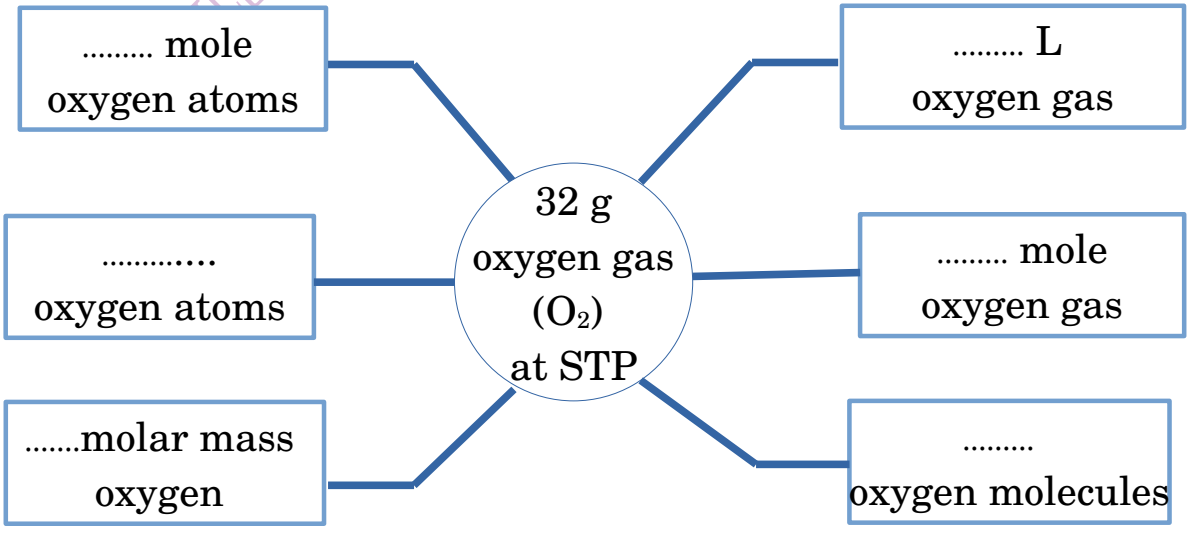
According to Avogadro's law, at constant temperature and pressure equal volumes of all gases contain equal number of molecules.

If temperature and pressure are fixed at 273 K and 1 atm respectively, it is known as STP (Standard Temperature and Pressure). At STP, one mole of any gas occupies a volume of 22.4 L. This is known as molar volume at STP.

$$\text{Number of moles of gases at STP} = \frac{\text{Given volume in litres}}{\text{Molar volume at STP}}$$

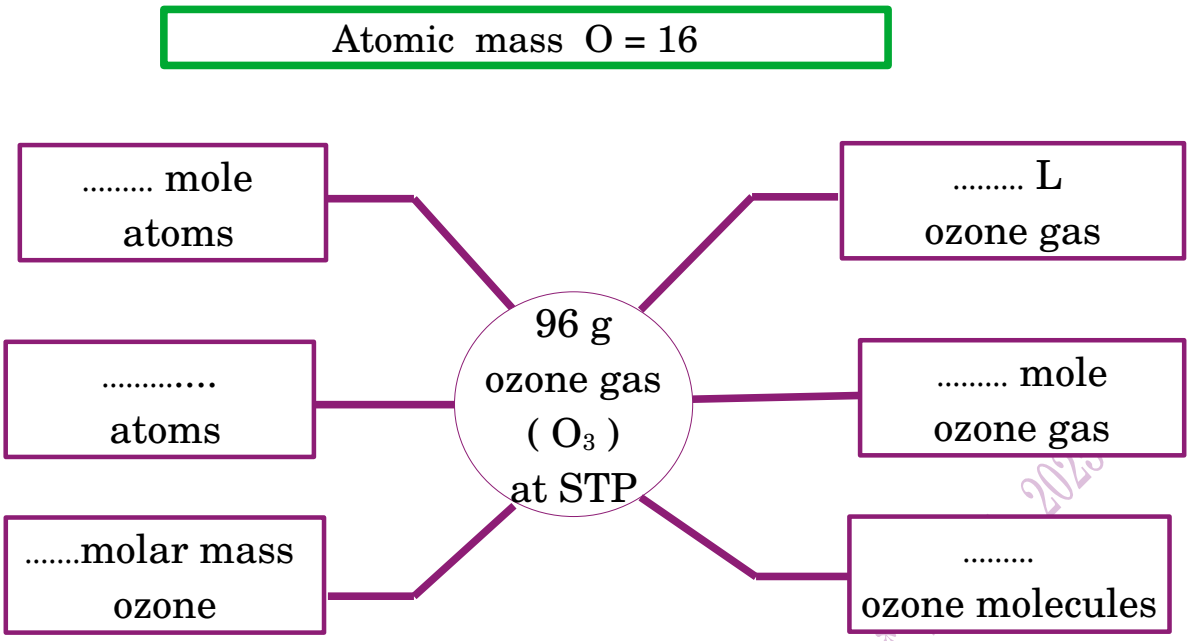
Gas	Molar mass	Given Mass	Number of moles	Volume (L)
O_2	.....	32 g	.....	22.4
NH_3	.....	170 g	.....	.....
CO_2	.....	.....	3	.....
NO_2	.....	.....	2	.....

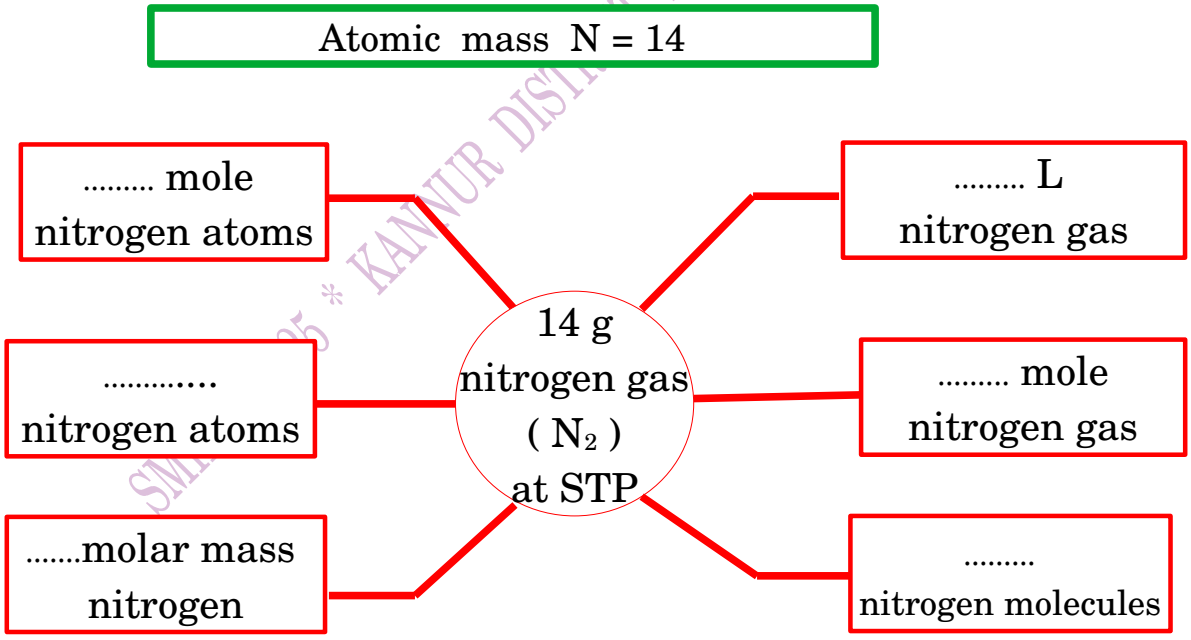
Atomic mass O = 16



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Ideal gas equation

$$PV = nRT$$

This is ideal gas equation. The gases which obey ideal gas equation at all temperature and pressure are known as ideal gases.

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Mole concept and chemical equations

Mass = mole x molar mass

molar mass $H_2 = 2$ g , $O_2 = 32$ g $H_2O = 18$ g

$2\ H_2 + O_2 \rightarrow 2\ H_2O$

Based on mole	2 mole H_2 + 1 mole O_2 → 2 mole H_2O
Based on mass	4 g H_2 + 32 g O_2 → 36 g H_2O

Based on mole	20 mole H_2 + ... mole O_2 → ..... mole H_2O
Based on mass	..... g H_2 + ... g O_2 → ..... H_2O

Based on mole	... mole H_2 + 3 mole O_2 → ..... mole H_2O
Based on mass	..... g H_2 + ... g O_2 → ..... H_2O

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1. Write the gas laws connected with the following situations

   a) The size of weather balloon goes on increasing as they rise above the sea level
   b) The container of liquid ammonia submerged in cold water for some time before opening
   c) The size of the air bubbles rising up from the bottom of an aquarium increases as they reach the surface of the water.

2. The volume of a definite mass of a gas at a pressure of 1atm is 44 L If the pressure increases to to 4atm what will be its new volume ?

3. Certain statements regarding gases are given below select the correct statement ?

   a) In gases molecules are far apart
   b) Energy of gas molecules are less
   c) Gas molecules have more freedom of movement

   A) Statements a,b,c are correct
   B) a and b correct c is wrong
   C) a and c are correct b is wrong
   D) All statements are wrong

4. a) 3 g hydrogen
   b) 16 g oxygen
   c) 28 g nitrogen
   d) 12 g helium

   Select the group of the above samples having same number of atoms

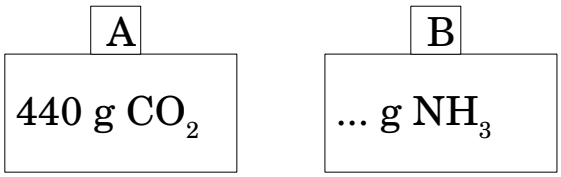
   i) a and b ii) a and d
   iii) c and d iv) b and c

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5. Two gases occupy equal volume at STP are given below.
(Atomic mass C = 12 , N = 14 , O = 16 , H = 1)



a) Calculate the number of molecules present in "B" ?
b) Find the mass of gas in "B" ?

6. You are asked to make 10 moles of glucose ($C_6H_{12}O_6$) into packets of 100 g each. (Atomic mass C = 12 , H = 1 , O = 16)

a) Calculate molecular mass of glucose?
b) What is the molar mass of glucose?
c) How many packets of glucose can be prepared?

7. Main component of LPG is butane.

a) Write molecular formula of butane.
b) Write chemical equation for the combustion of butane.
c) Calculate the mass of carbon dioxide formed due to the combustion of 11.6 kg butane?

8. The volume of a gas at 27° C and at 1 atm pressure is 100 mL. What will be its volume at 273 K temperature and 2 atm pressure?

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